

Business Requirements Document (BRD)

Retail Sales Performance Analytics & Intelligence Platform

Project Sponsor: Executive Leadership Team

Business Domain: Global Multi-Channel Retail

Document Version: 1.0 (Final)

Date: August 2026

1. Business Problem

OmniRetail Global operates across multiple international geographies (USA, UK, Germany, UAE, India, Pakistan) managing thousands of daily multi-category transactions. Despite strong historical gross revenue generation (\$32.13M across 50,000 recorded transactions), enterprise stakeholders face critical operational inefficiencies and strategic blind spots:

- **Suboptimal Customer Satisfaction:** Average customer feedback scores sit at **3.33 / 5.00**, indicating friction in product expectations, quality, or post-purchase experiences.
- **Fulfillment Bottlenecks:** Order fulfillment and delivery cycles average **8.0 days**, restricting competitiveness in high-velocity retail categories where speed drives repeat purchases.
- **Unoptimized Discounting Impact:** Across-the-board discounting averaging **12.3%** lacks dynamic elasticity governance, resulting in avoidable margin compression.
- **Siloed Regional Demand Visibility:** Fragmented demand signals across international hubs lead to inventory imbalances, stockouts, and inefficient capital allocation.

2. Project Scope

In-Scope:

- Integration and ingestion of historical transaction logs (Jan 2023 – Jun 2026) encompassing 50,000 records across 5 product categories (Apparel, Electronics, Home & Kitchen, Sports, Beauty) and 6 operating regions.
- Development of automated data transformation pipelines, dimensional modeling, and interactive executive analytics dashboards.
- Establishment of automated KPI monitoring for AOV, CSAT, delivery lead times, discount efficiency, and SKU-level profitability.
- Deployment of predictive models for demand sensing, customer sentiment categorization, and dynamic pricing elasticity.

Out-of-Scope:

- Direct refactoring of legacy Enterprise Resource Planning (ERP) or warehouse management transaction databases.
- In-store physical Point of Sale (POS) hardware upgrades or proprietary cashier terminal software replacements.

3. Business Objectives

- **Maximize Gross Margin Yield:** Rationalize discounting strategies by 2026 Q4 to recover at least 1.5% in annualized gross margin leakage.
- **Accelerate Fulfillment Cycles:** Streamline regional logistics partnerships to reduce average order-to-delivery lead time from 8.0 days to under 5.0 days.
- **Elevate Customer Sentiment:** Identify root causes of low customer ratings (≤ 2.0) and raise overall CSAT score from 3.33 to ≥ 4.25 within 6 months of rollout.
- **Optimize Inventory Turnover:** Implement regional demand-sensing algorithms to reduce excess safety stock holdings by 12% across international warehouses.

4. Stakeholders

Stakeholder Role	Department / Entity	Project Responsibility & Involvement
Executive Sponsor	Executive Leadership / C-Suite	Provides project funding, governance approval, and strategic direction.
Product Owner	Business Intelligence & Analytics	Manages requirement backlog, stakeholder alignment, and delivery roadmap.
Lead Data Analyst	Data Analytics & Engineering	Designs data schemas, data cleaning pipelines, and predictive models.
Operations Manager	Supply Chain & Logistics	Consults on fulfillment bottlenecks, shipping provider SLAs, and inventory workflows.
Commercial Director	Sales & Marketing	Defines pricing governance rules, discount thresholds, and promotional tracking.

5. Functional Requirements

Req ID	Feature / Module	Description
FR-01	Data Ingestion & ETL	The system shall automatically ingest, clean, and validate daily retail transaction files (CSV/JSON/SQL) covering customer demographics, order quantities, pricing, discounts, and ratings.
FR-02	Executive Dashboard	The platform shall provide role-based interactive dashboards displaying real-time KPIs (Total Revenue, AOV, CSAT, Fulfillment Duration, Discount Impact).
FR-03	Regional & Category Filtering	Users shall be able to filter and slice transactional analytics dynamically by country (USA, UK, Germany, UAE, India, Pakistan), product category, date range, and payment method.
FR-04		

Req ID	Feature / Module	Description
FR-05	Anomaly & Low-Rating Alerting	The system shall generate automated alerts for product SKUs or delivery corridors experiencing a sudden spike in customer ratings ≤ 2.0 or delivery times > 12 days.
	Discount Elasticity Simulator	The analytics engine shall provide a scenario simulation tool allowing commercial teams to test the revenue and margin impact of adjusting discount rates between 0% and 30%.

6. Non-Functional Requirements

Req ID	Category	Description
NFR-01	Performance & Latency	Dashboard queries and aggregated metric reports across 50,000+ historical records must load in under 2.5 seconds under standard concurrent user loads.
NFR-02	Security & Privacy	Customer identifiable information (Customer ID, demographic data) must be encrypted at rest and in transit, complying with global privacy frameworks (GDPR, CCPA).
NFR-03	Scalability	The data architecture must scale horizontally to support transaction volume growth up to 500,000 records annually without degradation in processing performance.
NFR-04	Availability & Reliability	The analytics platform shall maintain a 99.9% uptime availability SLA during standard global business operational hours.

7. Assumptions

- Historical transaction datasets provided (covering Jan 2023 to Jun 2026) are accurate, complete, and adequately sanitized for baseline modeling.
- Key department stakeholders (Supply Chain, Commercial, BI) will be available for bi-weekly user acceptance testing (UAT) and feedback sprints.
- Cloud infrastructure and database licenses necessary for hosting analytics pipelines and dashboard viewers are pre-approved.

8. Constraints

- **Timeline Constraint:** Full platform deployment and executive sign-off must be completed within a strict 12-week project window.
- **Integration Constraint:** Analytics pipelines must integrate seamlessly with existing enterprise data storage formats without requiring core database schema overhauls.

- **Resource Constraint:** Development relies on the designated cross-functional analytics team without additional external contractor augmentation.

9. Success Metrics

Metric Name	Baseline (Current)	Target (Post-Implementation)
Average Order Value (AOV)	\$642.53	≥ \$675.00 through cross-sell optimization
Customer Satisfaction (CSAT)	3.33 / 5.00	≥ 4.25 / 5.00
Fulfillment Lead Time	8.0 Days	< 5.0 Days
Discount Margin Leakage	12.3% average discount	Optimized to targeted 8.5% average discount